

Java Assignment - 02

Project: Smart Inventory & Billing Management System

Course: Core Java Programming

Assignment Type: Individual Project

Difficulty: Intermediate

Assignment Objective

The objective of this assignment is to evaluate your understanding of Core Java by developing a complete console-based Inventory and Billing Management System. The project focuses on applying object-oriented programming, collections, exception handling, file handling, and clean software design.

Learning Outcomes

- Object-Oriented Programming
- Classes & Objects
- Constructors
- Encapsulation
- Inheritance
- Polymorphism
- Abstraction
- Interfaces
- Collections Framework
- Exception Handling
- File Handling
- Serialization
- Lambda Expressions
- Stream API

Modules

1. Authentication

- Login using username and password.

2. Dashboard

- Product Management

- Customer Management
- Order Management
- Reports
- Logout

3. Product Management

- Add, Update, Delete Products
- Search by ID/Name
- View All Products
- Sort by Name, Price, Stock
- Low Stock Report

4. Customer Management

- Add, Update, Delete Customers
- Search Customers
- View Customer List

5. Order Management

- Create Orders
- Select Customer
- Add Products
- Quantity Management
- Stock Validation
- Invoice Generation

6. Reports

- Total Products
- Total Customers
- Total Orders
- Revenue
- Daily Sales
- Monthly Sales
- Low Stock Report
- Customer Purchase History

Technical Requirements

Students must demonstrate:

- Classes and Objects
- Constructors
- Encapsulation
- Inheritance
- Polymorphism
- Abstraction
- Interfaces
- Static Members
- Final Keyword
- Enums
- Java Collections
- Custom Exceptions
- File Handling
- Serialization
- Lambda Expressions
- Stream API

Suggested Package Structure

src/

- model/
- service/
- repository/
- exception/
- util/
- enums/
- interfaces/
- main/

Coding Standards

- Meaningful class and variable names
- Proper indentation
- Proper comments
- Follow OOP principles
- Separate UI from business logic
- Avoid duplicate code

Evaluation Criteria

Project Structure	10
Object-Oriented Programming	20
Collections	10
Exception Handling	10
File Handling & Serialization	10
Product Module	10
Customer Module	10
Order & Billing	15
Reports	5
Code Quality & Documentation	10
Total –	100 Marks